CARLETON UNIVERSITY
Department of Systems and Computer Engineering

SYSC 5104/4906G. METHODOLOGIES FOR DISCRETE EVENT MODELLING AND SIMULATION

Term project – Details

The goal of this document is to give you details about the development in the projects you have chosen and give you guidance about the steps to be followed in order to meet the requirements.

Deliverables (everybody)

The details of the deliverables will be discussed in detail for each of the projects. In every case, you should include a 15-20-page Term Paper explaining the results of your project, showing any results obtained, using IEEE, ACM or SCS format). It should include a brief description of each of the models used (as a conceptual model definition), besides the formal specification. The Term Paper should be organized into:

1. Introduction (brief description of the project, ideas, background and results obtained; it should be organized as i) Motivation (WHY are you doing this?); ii) Goals (WHAT are you doing it?); iii) Contributions (HOW did you do it?))
2. Background (any detailed information about background needed for the project)
3. Models defined (including an explanation of the models you built, any rules involved, and general description of the ideas implemented)
4. Simulation results (showing execution of the models in the simulation engine, and analyzing the results obtained)
5. Conclusions
6. References

Any source code should run cleanly in the simulation tools, using similar guidelines to the ones used in Assignment 1 and 2. In order to ensure proper execution, try to execute the same steps explained in the assignments before delivering the software.

DEADLINE: April 22, 2025 (earlier submissions are accepted).

Formatting instructions for the Term Papers:
http://www.acm.org/sigs/publications/proceedings-templates
http://www.scs.org/PDFs/formattingkit.pdf

(a good example on organization of the Term Paper can be found in:
http://www.sce.carleton.ca/faculty/wainer/papers/SIW-05s-Battle.pdf)

The authors of the best projects will be invited to be extended and submitted for publication, if the students are interested. The final projects will be published on-line in the course webpage for perusal of interested modelers.

The source code should run cleanly. Include shell scripts to run the model, explaining how to run them, which input files are needed for each case. Include the document of the final report. Follow the guidelines for Assignment 1 and 2 for submitting. Use meaningful names everywhere ("FinalProject" or "MyName" are examples of names you should not be using; instead, names like "WirelessNetworkSimulator.zip" are

accepted).

The reports will be analyzed as follows:
. The files submitted will be recompiled using your files, generating the simulator.
. The examples will be run using the scripts you provided.
. After these basic steps are carried out, different changes will be studied, in order to find possible errors.

<u>In order to ensure proper execution, try to execute these same steps by yourselves, before delivering the software.</u>

Include 'read.me' files explaining the detailed behavior for each one of the scripts provided. Include the electronic version of your final reports in the zip file.

Below you will find generic descriptions of the topics; if there are missing details, please consult.

Details of the proposed projects

I. Building Advanced Cell-DEVS models

We used Cadmium to develop different models in class. The goal of this project is to define advanced examples of complex physical systems using Cell-DEVS and Cadmium v2. The initial version of the project should be an extended version of Assignment 2. The definitive version should be an improved version, including phenomena not originally included in the papers you used to define the models, and using asymmetric Cell-DEVS models. For all spatial models involving geographical information, you should integrate the model with QGIS, as shown in the example below.

i) Read the documentation Asymmetric Cell-DEVS in the article and video uploaded to Brightspace
ii) Define the model using Asymmetric Cell-DEVS and varied topologies (ideally extracted from QGIS)
iii) Extend the model and include your own examples
iv) Enhance the model described in Assign 2 with new rules, extending the original ones
v) Include videos of visualizations

Detailed deliverables:
Submission: similar to Assignment 1 and 2 (follow the To-do List)

- Definition of different examples with different input values.
- Execution results of these examples
- Final Report

Information:

Wildfire model with integration with QGIS:

https://github.com/murf85/wildfire

. Dengue models

- Use the Dengue models below as a starting point (CD++ models)
- Check the articles uploaded
- Check the information below
- Build your own models (discuss details with Prof. Wainer)

https://github.com/SimulationEverywhere-Models/CDPP-DengueModels
https://github.com/Abdalla-00/Dengue-Fever-Spread-Model